SOLVING QUADRATIC EQUATIONS

BASIC ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Rational Expression

Standard Form

Quadratic Formula

RATIONAL EXPRESSION

RATIONAL EXPRESSION

Simplify $\frac{21}{45}$ by reducing to **lowest terms**

A mathematical expression that can be represented in a **fractional form**

Simplify $\frac{3x+6}{x^2+4x+4}$

by reducing to **lowest terms**

EXERCISE

Compute $\frac{4}{3} \cdot \frac{2}{5}$

Compute $\dfrac{\frac{4}{5}}{\frac{2}{3}}$

Compute $\dfrac{\dfrac{x^2+x}{x+4}}{\dfrac{x+1}{x^2-16}}$

Subtract $\frac{7}{6} - \frac{4}{15}$

Add $\frac{3}{2x+2} + \frac{5}{x^2-1}$

STANDARD FORM

STANDARD FORM

A quadratic equation is a **second**-degree polynomial equation

$$ax^2 + bx + c = 0$$

Find all **real solutions** for the equation

$$y^2 = 18 - 7y$$

Find all **real solutions** for the equation

$$w^2 = 121$$

Find all **real solutions** for the equation

$$x(x+2)=7$$

QUADRATIC FORMULA

QUADRATIC FORMULA

The **solutions** to the quadratic equation

are given by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Find all **real solutions** for the equation

$$x(x+2)=7$$

Find all **real solutions** for the equation

$$\frac{1}{2}y^2 = \frac{1}{3}y - 2$$

SEATWORK